

Aim: To understand their respective role in networks/internet.

Role of NIC Cards (Network Interface Cards): A **Network Interface Card (NIC)** is the hardware component that connects a computer or device to a network. It acts as the interface between the system and the communication medium.

1. Physical Connection: The NIC provides a physical port such as an RJ45 socket for wired networks or an antenna for wireless networks. It acts as the entry and exit point for all network data.

2. Data Conversion: The NIC converts digital data from the computer into electrical or radio signals for transmission. At the receiving end, it converts incoming signals back into digital data that the computer can understand.

3. Addressing: Each NIC has a unique **MAC address**, which is used to identify the device on a local network. This address helps switches deliver data frames to the correct destination.

4. Frame Processing: The NIC breaks data into frames before transmission and performs error detection using methods like CRC. It also filters incoming frames by checking whether the destination MAC address matches its own.

5. Speed Negotiation and Duplex Control: NICs automatically negotiate speed (10, 100, or 1000 Mbps) with connected devices and support half-duplex or full-duplex communication.

Importance: Without a NIC, a computer cannot communicate over a network. It acts as the network communication engine of the device.

Role of Ethernet, Fast Ethernet, and Gigabit Ethernet: Ethernet standards define how data is transmitted over a wired network, including speed, encoding and access methods.

1. Ethernet (10 Mbps): Ethernet provides the basic rules for data transmission in a LAN. It uses the CSMA/CD method to detect and manage collisions. It is suitable for small and low-speed networks.

2. Fast Ethernet (100 Mbps): Fast Ethernet increases the data rate to 100 Mbps while keeping the same frame structure as Ethernet. It reduces transmission time and supports faster communication. It requires better quality cables like Cat5 or Cat5e.

3. Gigabit Ethernet (1000 Mbps): Gigabit Ethernet offers very high speed and is used in modern networks for video streaming, gaming, servers and large data transfers. It uses all four twisted pairs in the cable and supports full-duplex transmission.

4. Summary: Ethernet standards ensure compatibility between devices by defining transmission speed, cabling, encoding and data handling rules.

Role of RJ45 Pins (Connector Pinout): An **RJ45 connector** is used at the end of Ethernet cables to connect devices like NICs, switches and routers. It has **8 pins**, each connected to a wire inside the cable.

1. Signal Transmission: Pins 1 and 2 are used for transmitting data, while pins 3 and 6 are used for receiving data in standard Ethernet. In Gigabit Ethernet, all eight pins are used for data transmission.

2. Wiring Standards: RJ45 connectors follow wiring standards such as **T568A** and **T568B**. These standards ensure correct wire placement and prevent signal interference.

3. Maintaining Signal Quality: Correct pin alignment keeps twisted pairs intact, reducing crosstalk and electromagnetic interference.

4. Importance: If RJ45 pins are incorrectly wired, communication may fail or network speed may reduce significantly.

Role of Twisted-Pair Cable: A **twisted-pair cable** is the physical medium used to carry data signals in wired networks.

1.Signal Transmission: It carries electrical signals representing binary data (0s and 1s) between network devices.

2.Noise Reduction: Each pair of wires is twisted together to cancel electromagnetic interference and reduce crosstalk between adjacent wires.

3.Speed Support: Different cable categories support different speeds:

- Cat5: up to 100 Mbps
- Cat5e: up to 1 Gbps
- Cat6 / Cat6a: up to 10 Gbps

4.Distance and Reliability: Twisted-pair cables can carry signals reliably up to 100 meters and support full-duplex communication.

5.Why It Is Used: Twisted-pair cables are cost-effective, flexible, easy to install and capable of high-speed data transmission.

How All Components Work Together:

- The **NIC** creates data frames and converts them into electrical signals.
- **Ethernet standards** define how fast and in what format the data is transmitted.
- The **RJ45 connector** organizes signals through its pins and connects the cable to devices.
- The **twisted-pair cable** carries the signals to another device.
- At the receiving end, the process is reversed and the data reaches the destination computer.